

Andrews Acheampong

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Kumasi, Ashanti - AK-144-8245, Ghana

RESEARCH INTERESTS

I am interested in scalable Bayesian inference and distributed MCMC, and how such methodologies can be developed and applied to challenging problems in molecular biology and genomics.

EDUCATION

• Kwame Nkrumah University of Science and Technology

January 2026 - Present

MPhil. Biodata Analytics and Computational Genomics

Kumasi, Ghana

◦ First Semester Results:

* WA: 72.33/100

* GPA Equivalent: 3.53/4.00

* Credits Obtained: 18

• Kwame Nkrumah University of Science and Technology

January 2021 - August 2024

Bachelor of Science in Statistics

Kumasi, Ghana

◦ CWA: 74.49/100

◦ GPA Equivalent: 3.82/4.00

◦ Total Credits Obtained: 154

RESEARCH EXPERIENCE AND LABORATORY EXPERIENCE

• Genetic and Statistical Foundations of Coalescent Theory and Its Applications in Population Genetics

Graduate Term Paper

July 2026

◦ Examined coalescent theory through the Wright-Fisher model, stochastic processes, and backward-time population genetic inference.

◦ Applied geometric, exponential, and Poisson models alongside Watterson's estimator, nucleotide diversity, the site-frequency spectrum, and Tajima's (D).

◦ Investigated statistical inference of effective population size, demographic history, migration, natural selection, and genetic ancestry.

• Kumasi Veterinary Laboratory, Molecular Department

May 4-22, 2026

Wet Laboratory Intern, Molecular Diagnostics

Kumasi, Ghana

◦ Completed supervised molecular diagnostics training involving DNA and RNA extraction from laboratory samples.

◦ Prepared PCR master mixes and carried out PCR-based viral detection workflows under supervision.

◦ Interpreted amplification outputs with supervisor guidance, including the expected exponential amplification curve pattern for target-positive reactions.

◦ Practiced contamination-aware sample handling, pipetting accuracy, laboratory documentation, and careful reaction setup.

• Predictive Analytics for Kidney Disease Progression

Undergraduate Thesis

◦ Investigated the impact of hemoglobin, serum creatinine, hypertension, and random blood glucose on CKD risk.

◦ Developed a predictive model for Chronic Kidney Disease (CKD) using logistic regression.

◦ Achieved 97% accuracy and a Kappa value of 0.94 through 10-fold cross-validation.

ACADEMIC AND TEACHING APPOINTMENTS

• Department of Statistics and Actuarial Science, KNUST

October 2024 - October 2025

Research and Teaching Assistant

Kumasi, Ghana

◦ Conduct tutorial sessions for Probability Distributions and assisting undergraduate students in problem-solving.

◦ Design and prepare homework activities for students in Probability and Measure Theory, and Probability and Statistical Inference.

◦ Assist faculty members in research projects focused on data science and machine learning, providing guidance to final-year undergraduate students.

• Self-Employed

September 2024 - July 2025

Instructor

Remote

◦ Instructing students the basics of data analysis using R.

◦ Developed and delivered course materials covering data manipulation, visualization, and statistical analysis in R.

◦ Provided hands-on experience with real-world datasets and case studies focused on medical applications.

BIOINFORMATICS AND COMPUTATIONAL PROJECTS

• Rhinovirus Classification Web App: [Web application]

January 2025

Tools: [Python, Flask, Jinja2, SQLite, Matplotlib, Biopython, HTML, JavaScript, CSS]



- Developed a web application for analyzing and visualizing FASTA sequence data, supporting Rhinovirus classification.
- Integrated phylogenetic analysis methods, enabling users to upload sequences, perform analyses, and interpret results through interactive visualizations.

• Analysing and Interpreting Genomics Datasets: [Genomics Training Repository]

Tools: [Linux, Bash, HTML, RStudio, Nextflow concepts, NGS workflows]



- Built a practical genomics repository covering NGS concepts, genomics file formats, sequence quality control, mapping, and variant-calling workflows.
- Documented exposure to workflow management with Nextflow and viralrecon, strengthening preparation for scalable and reproducible bioinformatics pipelines.
- Connected Linux command-line analysis with RStudio-based visualization for interpreting genomic variant outputs.

• Bioinformatics for Biologists: [Command-Line Bioinformatics]

Tools: [Bash, Linux, Biological Data Workflows]



- Completed shell-based bioinformatics exercises focused on command-line workflows and biological data analysis foundations.
- Strengthened practical skills in reproducible directory organization, terminal-based analysis, and biological data handling.

• Automated Data Extraction: [Scripting]

April 2022

Tools: [Selenium, Pandas, Python]



- Implemented a web scraping script using Selenium and XPath to extract structured data from dynamic web pages.
- Processed and transformed scraped data using Pandas, ensuring clean and structured datasets for analysis.

PRESENTATIONS AND INVITED LECTURES

• Keynote Address: Data-Driven Decision – Building the Future We Want with Statistics

February 12, 2025

Ghana Association of Statistics Students (GASS), KNUST

- Delivered a keynote speech on the role of statistics in shaping data-driven decision-making.
- Highlighted the impact of statistical methods in policy development, business intelligence, and research.

• Effective Teaching Strategies

January 20, 2025

Department of Statistics and Actuarial Science, KNUST

- Delivered a structured presentation to newly admitted lecturers on effective teaching and assessment strategies.
- Explained the two main types of teaching strategies: teacher-centered and student-centered approaches.

• Introduction to Git and GitHub for Research Collaboration

November 19-21, 2024

Department of Statistics and Actuarial Science, KNUST

- Conducted a hands-on workshop for graduate and research assistants on version control using Git and GitHub.
- Explained key concepts such as repositories, branching, commits, pull requests, and merging.

COMPUTING SKILLS

- **Programming Languages:** Python, R, SQL
- **Web Technologies:** Flask, HTML, CSS, JavaScript, Bootstrap
- **DevOps & Version Control:** Git, GitHub, Conda, Bash scripting

CERTIFICATIONS

- **Basic Molecular Biology Module 3** August 2025
- **An Introduction to Linux, Bash Scripting, and R** March 2025
- **Full Stack Software Engineering** February 2025
- **Basic Molecular Biology Module 1** September 2024
- **Data Science with R** April 2024
- **Nanodegree Data Analysis** October 2022
- **Data Science Bootcamp** April 2022

REFERENCES

Available upon request.